Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

We compare gravity, drag, constant net force and a gravity–drag–Magnus model. M3 estimates constant acceleration from early positions; M4 fits a constant drag coefficient and a lift coefficient proportional to the spin parameter.

Three recorded penalties provide 39 tracking frames. After conversion to penalty-mark coordinates, three selected positions per event enter 12 bounded inversions, compared with later samples. J03WMX and J03WR9 are analyzed as placement shots and J03WOH as a power shot, comparing speed, aerodynamic forces and prediction error. A separate curved-shot calculation examines rotational damping. Section 5 uses the competition positions for individual perturbations, position-constrained crossings and 100,000 Monte Carlo trajectories.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and rotation can produce a sideways or vertical aerodynamic force. Bray and Kerwin combined aerodynamic forces with equations of motion and compared the resulting trajectories with experimental free kicks [5]. Myers and Mitchell analyzed three-dimensional spinning-ball motion and the influence of drag and spin [10]; Maponi and Piombin developed a numerical framework that also accommodates wind [8].

The aerodynamic behavior is also affected by the ball's surface. Wind-tunnel measurements by Asai et al. examined the relationship between flow conditions and soccer-ball forces [11], while Paudel and Yan's simulations compared a football with realistic seam geometry against a smooth sphere [9]. These studies motivate a spherical geometric approximation together with an examination of how aerodynamic coefficients affect the predicted flight.

Measurements of real flights provide a basis for evaluating such models. Griffiths et al. tracked the three-dimensional motion and spin of a marked football in field conditions [3]. Goff and Carré used camera-recorded trajectories to estimate aerodynamic coefficients [15]; subsequent work by Goff et al. combined measurements from 29 flights to construct speed-dependent drag and lift curves [4].

In this problem, the initial position and two early positions are given, whereas the initial velocity and spin are not directly specified. We therefore use the equations of motion to find states compatible with the early observations and calculate their goal-plane crossings. The DFL open dataset combines match events and optical tracking data for professional matches [6, 7], allowing predicted crossings to be compared with later recorded positions.

1.2 Problem Restatement

We study a ball travelling from the penalty mark to a goal 7.32 m wide and 2.44 m high, 11 m away. The motion and prediction problem can be organized into three stages:

1. **Ball–foot contact.** Contact establishes launch velocity and spin. These initial conditions relate to the unknown motion reconstructed in Section 4.

2. **Airborne flight.** Gravity, drag and Magnus lift develop the trajectory according to the governing equations established in Section 3.

3. **Goal arrival and shot outcome.** The goal-plane crossing connects the flight to the shot result. Variation in initial motion, aerodynamic parameters and environmental conditions can produce a distribution of crossing locations; Section 5 examines these sensitivities and joint perturbations.

We estimate initial states from three real penalties and compare placement and power shots with later recorded positions. Figure 1 summarizes this physical and computational sequence.

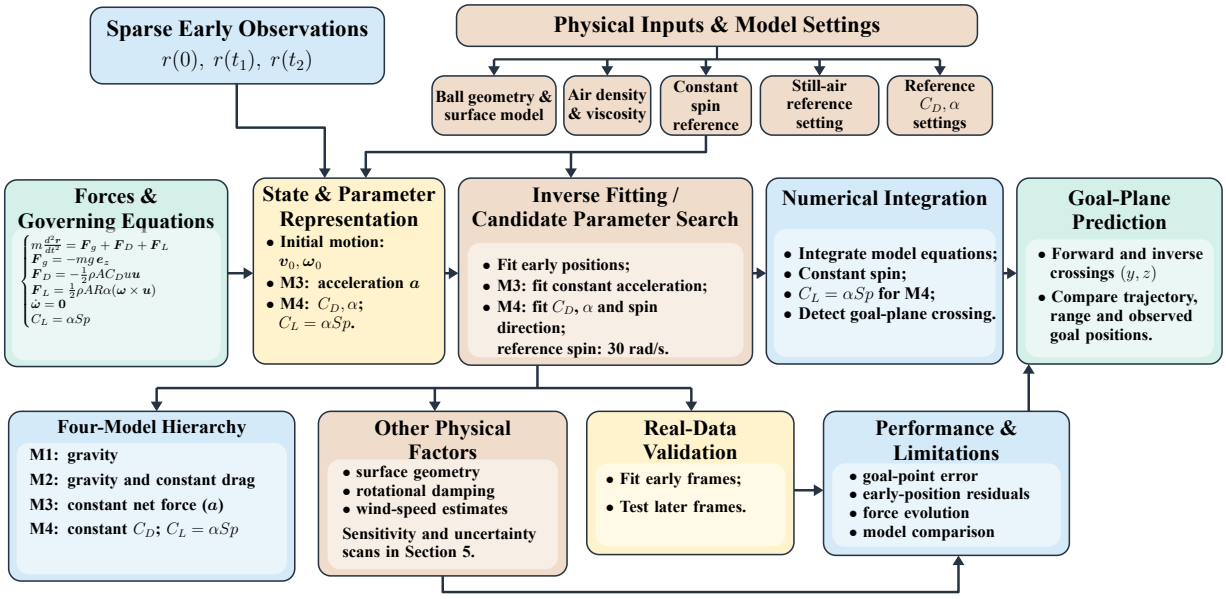


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

- **Contact and model start.** Real inversions define $t = 0$ at the selected starting tracking frame; contact is represented through the initial motion.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [8, 5, 10]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, aerodynamic drag and spin-induced lift act during free flight [8, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.
- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly vary-

ing gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.

- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknowns.** Three specified real-penalty positions enter a bounded deterministic fit, and subsequent positions supply the comparison. Released Z values are interpreted as metres and centre height. Section 4 states the search ranges for velocity, spin and force corrections; Section 5 uses the competition-position example to examine parameter effects on crossing distributions.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [8]; the air density follows Asai et al. [11]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. The three match penalties use the same declared reference settings.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

The local horizontal origin is the nominal penalty mark, with x towards the goal, y lateral and z upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m. Section 4.1 converts the measured pitch-centred coordinates.

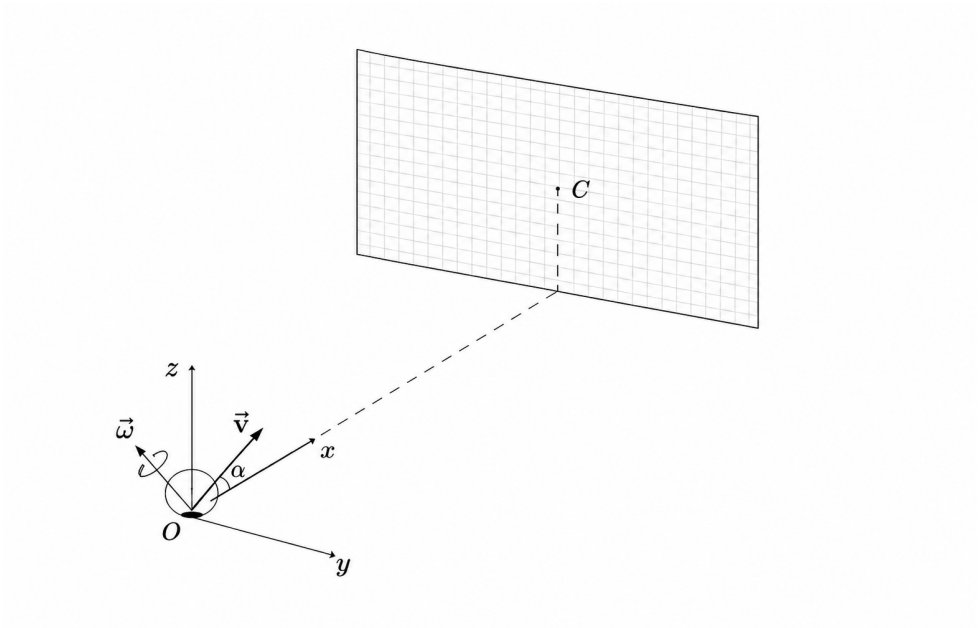


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

A clean opening crossing requires $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$. Descending centre height $z = R$ defines ground contact. Every model uses these same event definitions.

3.2 Force Analysis

3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. For the reference parameters in the table, buoyancy is about 1.56% of the ball's weight and changes the downward acceleration from g to $g(1 - \rho V_b/m)$.

The 1.56% buoyancy scale is a small vertical correction; all four main comparisons use gravity, drag and Magnus lift as defined above.

Figure 3 illustrates the relative airflow in the ball frame and the corresponding force directions. The right-hand example takes the spin axis out of the page; the lift direction follows the cross product.

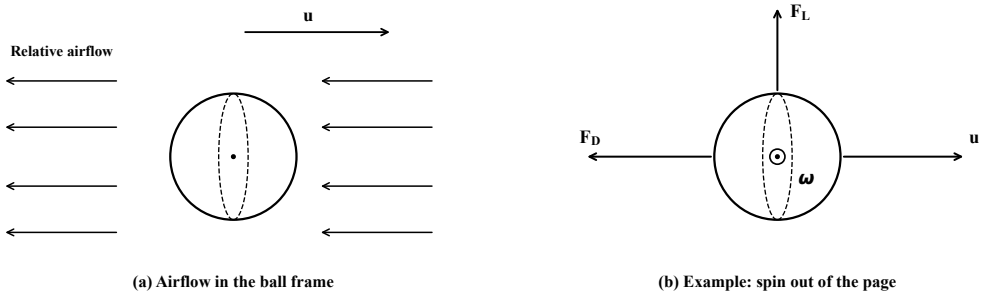


Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

3.2.2 Aerodynamic Drag

With relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$, $u = \|\boldsymbol{u}\|$ and $A = \pi R^2$, drag is [15, 12]

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A C_D u \boldsymbol{u}. \tag{3}$$

The drag coefficient C_D is constant during each flight.

3.2.3 Magnus Lift

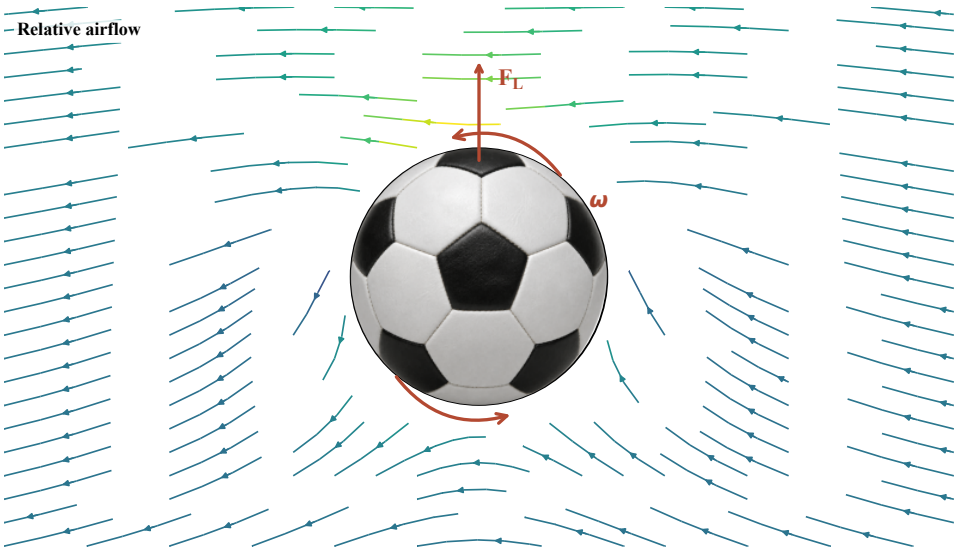


Figure 4: Idealized flow around a spinning football.

Let $Sp = R\|\boldsymbol{\omega}\|/u$ and $C_L = \alpha Sp$. Resolving the transverse component yields

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{4}$$

The force vanishes at zero spin or zero relative velocity; reversing cross-product order requires reversing the sign. The forces on the football are summarized in the following diagram.

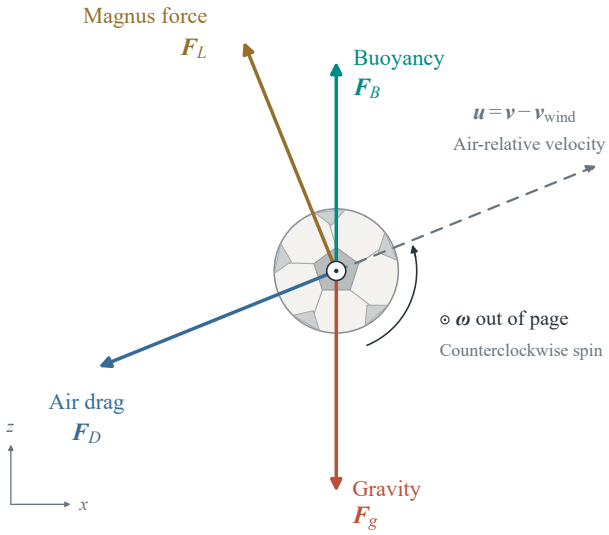


Figure 5: Forces on the football: gravity, buoyancy, drag and Magnus force, with arrows showing their directions for the illustrated state.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b \boldsymbol{\omega}. \tag{5}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{6}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{7}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

4 Model Solving

We infer initial motion from three recorded penalties and compare the flight and prediction of placement and power shots.

Table 2: Models and inverse quantities

Model	Assumption	Inverse quantities
M1	gravity	velocity (3)
M2	gravity and drag	velocity, drag coefficient (4)
M3	constant net force	velocity, acceleration (6)
M4	gravity, drag, Magnus force	velocity, spin direction, C_D, α (7)

M3 represents the combined net force. M4 uses constant C_D and α during each flight with $C_L = \alpha Sp$. Forward reference values are $C_D^* = 0.2415$ and $\alpha^* = 1.5251$ at a published reference condition [12].

4.1 Inverse Prediction from Three Real Penalties

4.1.1 Measurements, coordinate system and input times

The Sportec Open DFL data associate actual match events with TRACAB Gen5 multi-camera tracking at 25 Hz [6, 7]. We use penalties by Dejan Ljubicic (J03WMX), Dawid Igor Kownacki (J03WOH), and Philipp Klement (J03WR9). The supplied CSV retains 39 source frames, respectively 15, 12 and 12. Positions are used as published; no added observation-noise model is introduced.

The source horizontal origin is the pitch centre. All three shots travel towards $X = 52.5$ m. To use the competition's penalty-mark coordinates, we apply

$$x = X - 41.5, \qquad y = Y, \qquad z = Z_{\text{original}}, \qquad x_g = 11 \text{ m}. \tag{8}$$

The nominal penalty mark is the horizontal origin, so each actual first observation retains its measured offset. Released Z values are interpreted as metres and ball-centre height. Mass, radius and air properties use the model reference settings in Section 2; the supplied source leaves per-ball mass and radius unreported.

Each inversion defines $t = 0$ at its selected source frame. J03WMX uses samples 3, 4 and 5 of the original sequence; J03WOH and J03WR9 use samples 2, 3 and 4. All three fitting times are 0, 0.04 and 0.08 s. Their time origins on the original recording are 0.08, 0.04 and 0.04 s, respectively. Table 3 gives the measured starting positions. Earlier records, repeated frames and post-goal records remain in the raw CSV. Trajectory comparison begins at the selected start, with subsequent samples used only for comparison up to the observed goal crossing. Frame-interpolated crossings are marked separately from measured frames.

Table 3: Frame counts, fitting times and measured local starting positions.

Event	Frames	Fit times (s)	$\boldsymbol{r}_0$ (m)
J03WMX	15	0.00,0.04,0.08	$(3.04, 0.55, 0.13)$
J03WOH	12	0.00,0.04,0.08	$(0.77, 0.08, 0.24)$
J03WR9	12	0.00,0.04,0.08	$(1.47, 0.49, 0.23)$

4.1.2 Bounded inversion and selection

M3 determines constant acceleration from three positions:

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \tfrac{1}{2}\boldsymbol{a}t^2. \tag{9}$$

M4 minimizes the position error over velocity, spin direction, C_D and α. Six deterministic initial guesses are used within the ranges in Table 4; the minimum-residual solution is retained.

Table 4: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi]$, $[-\pi/2, \pi/2]$ rad
$C_D/C_D^*, \alpha/\alpha^*$	$[0.5, 1.5]$

Only the product $\alpha\boldsymbol{\omega}$ enters Magnus force, so spin magnitude and α are not separately identified by positions. A reference spin magnitude of 30 rad/s is assumed. Seven fitted quantities and six later position coordinates leave multiple possible solutions.

4.2 Case Analysis and Validation

The recorded penalties fall into two kick types: J03WMX and J03WR9 are placement shots, and J03WOH is a power shot. Figure 6 compares their spatial trajectories. Open circles indicate fitting inputs, black lines and points denote tracking, colored lines show the models, and crosses mark measured or interpolated goal crossings. Predictions stop at first ground contact.

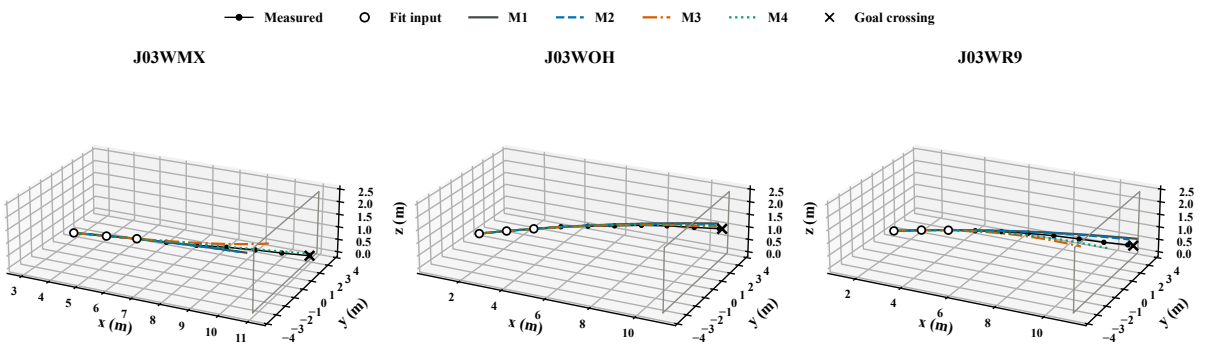


Figure 6: Three recorded penalty trajectories and model predictions. Each inversion uses positions at 0, 0.04 and 0.08 s after the selected start.

4.2.1 Case 1: Placement Shots

J03WMX and J03WR9 both travel close to the ground. M4 estimates their initial speeds at the selected frames as 29.20 and 29.33 m/s, respectively. For J03WMX, M4 reaches the goal plane with a 0.360 m error; M1 and M2 hit the ground and M3 does not reach the goal within the calculation horizon. For J03WR9, M1 and M2 have goal-plane errors of 0.550 and 0.541 m, while M3 and M4 predict ground contact before the goal.

A low trajectory is sensitive to vertical forces. Gravity pulls downward, while the vertical Magnus component can raise or lower the path. M1 omits aerodynamic forces, and M2 adds drag without transverse lift. M3 fits a constant acceleration to the early points, but its continuation can deviate as acceleration evolves. M4 allows aerodynamic forces to change with velocity and spin direction; it reaches the goal more accurately for J03WMX but predicts early ground contact for J03WR9. Early tracking and the fitted parameters both influence these differences.

4.2.2 Case 2: Power Shot

M4 estimates the J03WOH speed at the selected start as 32.05 m/s, higher than for the two placement examples. Figure 7 shows its spatial path and x, y, z against time. The first three positions are fitted; later black points assess the continuation.

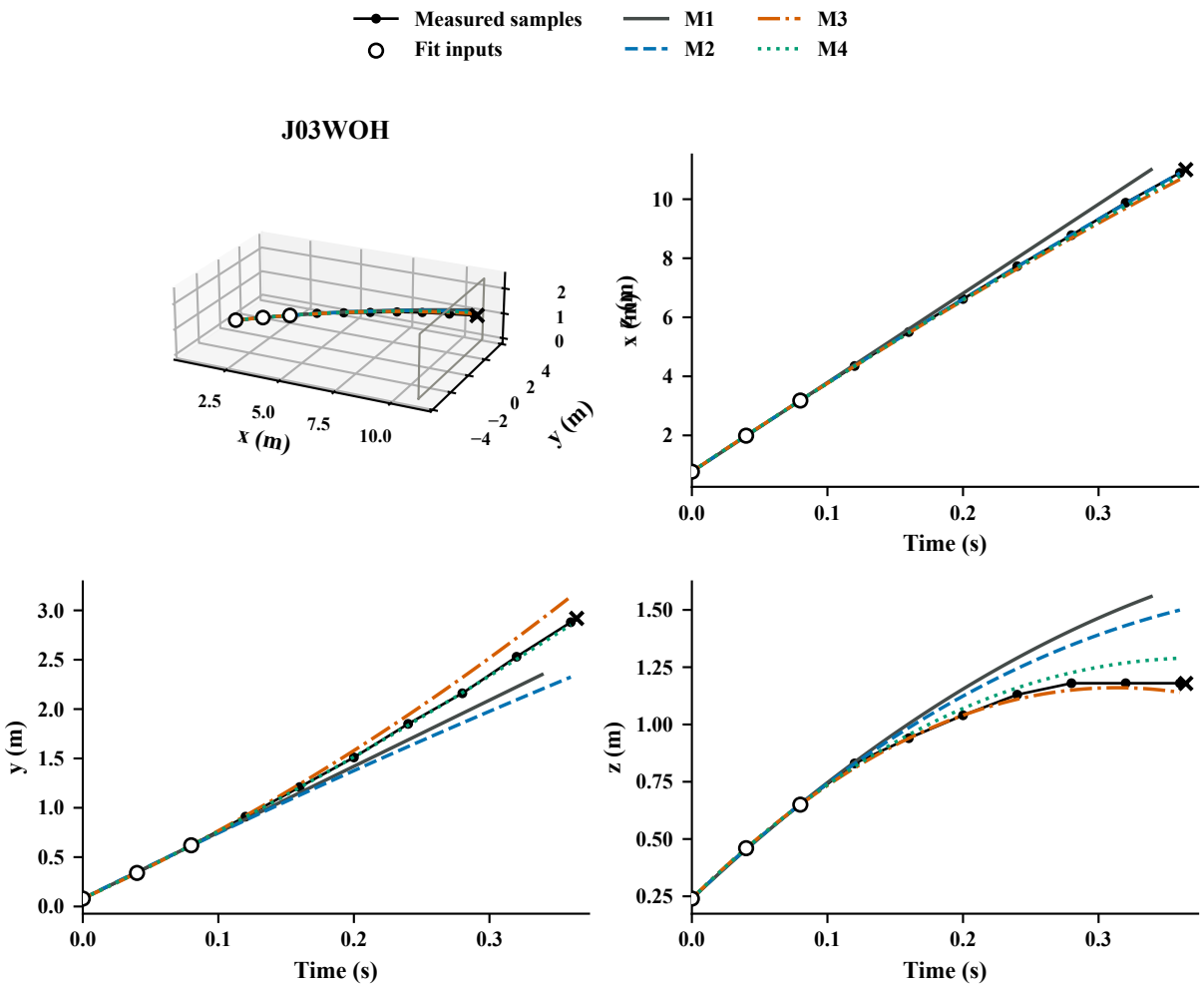


Figure 7: J03WOH power-shot trajectory and coordinates against time. Open circles are fitting inputs and black points are measured samples.

$x(t)$ shows forward deceleration; $y(t)$ and $z(t)$ show lateral and vertical curvature. M1, M2, M3 and M4 have goal-plane errors of 0.684, 0.657, 0.374 and 0.113 m, respectively. M1 uses gravity alone and M2 adds drag; their horizontal directions are mainly set by initial velocity. M3 continues the constant acceleration determined from the three inputs. M4 uses drag and Magnus force to describe evolving curvature and agrees more closely with the later points in this shot. Errors compare each model's own (y, z) crossing with the measured crossing.

Physical picture of speed and aerodynamic forces. Figure 8 compares M4 speed and force magnitudes. Drag scales with relative speed squared: the fitted initial drag is about 7.58 N for J03WOH and 7.05 and 7.11 N for the placement shots, with coefficient values also affecting the comparison. As the ball slows, drag decreases. Magnus force depends on the cross product of spin and relative velocity: spin direction determines whether its effect is chiefly lateral or vertical, and its magnitude changes with the motion. The displayed speeds and forces are computed from the M4 inversions, using a reference spin magnitude of 30 rad/s.

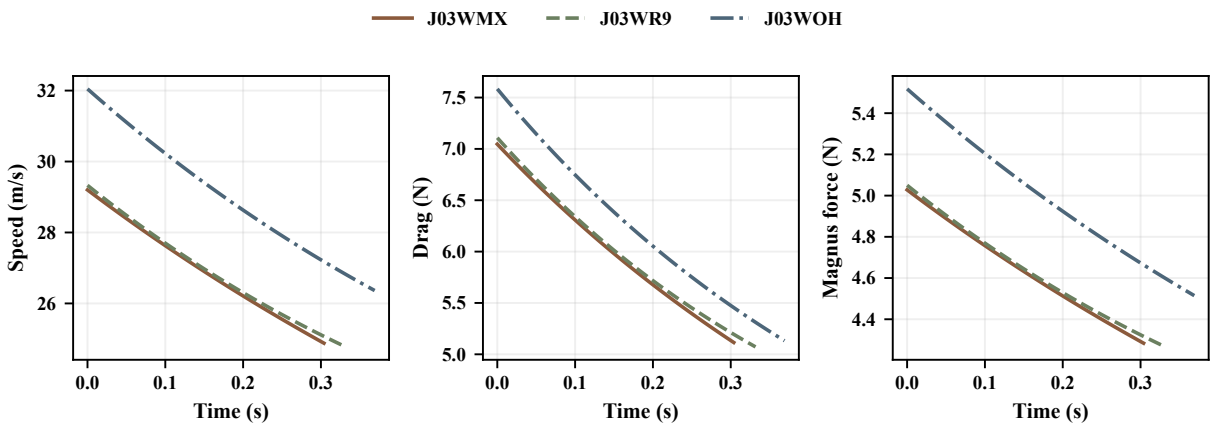


Figure 8: M4 speed, drag and Magnus force for the three recorded penalties. Curves end at first model-predicted ground contact or goal-plane crossing.

5 Sensitivity Analysis

We retain the individual-parameter, early-position-constrained and Monte Carlo structure for the current M4 equations. The reference observations are $(0, 0, 0.11)$ m at 0 s, $(1.50, 0.45, 0.44)$ m at 0.05 s, and $(2.96, 0.86, 0.74)$ m at 0.10 s. Mass and radius are fixed at 0.43 kg and 0.11 m.

The fitted reference uses $\boldsymbol{v}_0 = (30.3784, 9.4074, 6.8974)$ m/s, $\boldsymbol{\omega}_0 = (19.7434, 1.2060, -22.5553)$ rad/s, $C_D = 0.3623$ and $\alpha = 2.2877$. The crossing is $(2.6159, 1.9526)$ m at 0.398967 s. The 30 rad/s spin magnitude is a reference setting; the maximum early-coordinate residual is 0.000784 m and both coefficients reach the fitted upper bounds.

5.1 Effects of individual parameters

One parameter changes at a time: speed by $\pm 5\%$; spin magnitude, drag coefficient, lift parameter and air density by $\pm 10\%$; velocity and spin-axis angles by $\pm 2^\circ$. Each uses 41 values, totaling 369 flights. The table compares $\pm 1^\circ$ angles and $\pm 5\%$ scalar changes.

Table 5: Goal-plane displacement and arrival-time change under one-factor perturbations.

Parameter	Perturbation	max D_g (cm)	max $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag multiplier	$\pm 5\%$	0.620	2.274
Lift multiplier	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

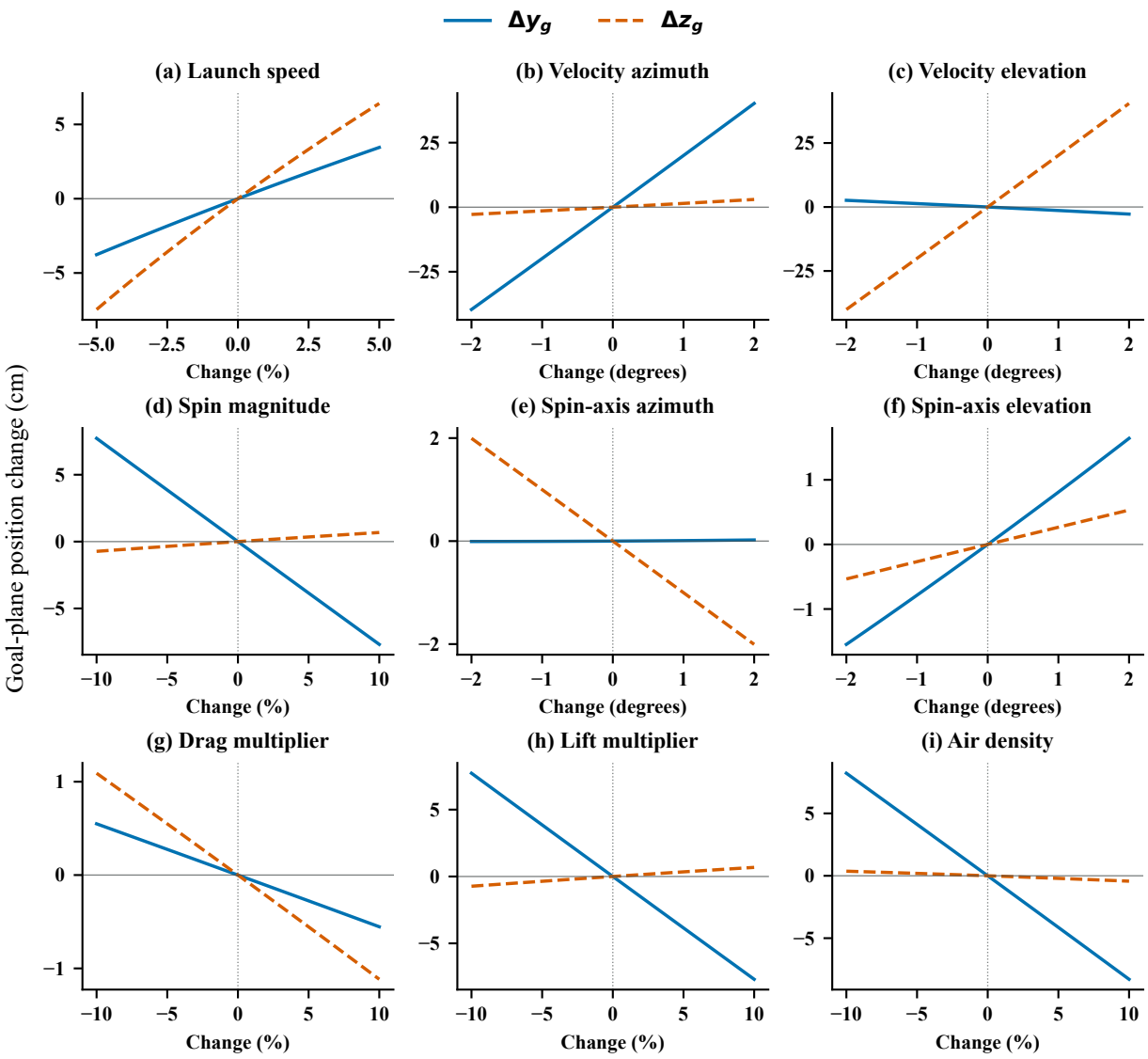


Figure 9: Goal-plane coordinate responses to the nine individual parameters.

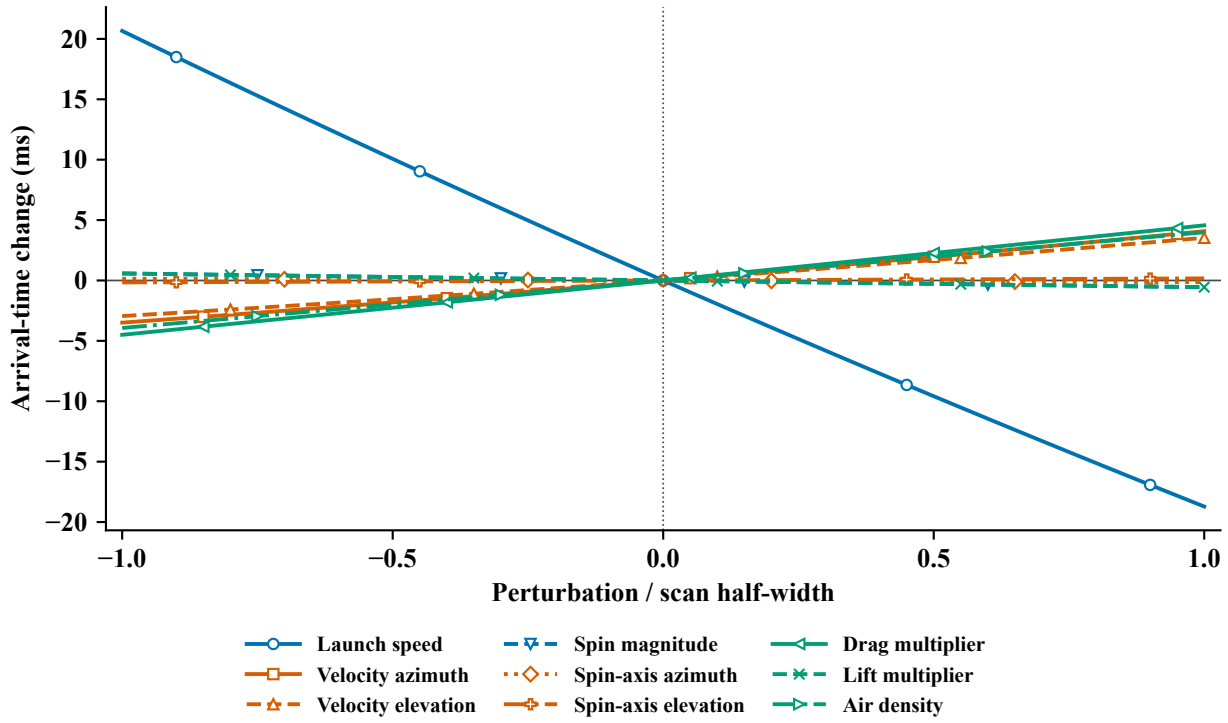


Figure 10: Arrival-time response, with perturbations normalized by each scan half-width.

Spin magnitude and α enter Magnus force as a product, so changes in opposite directions can compensate.

5.2 Crossing variation under early-position constraints

We scan a 17×17 grid of the two coefficient multipliers over $[1.2, 1.8]^2$, refitting velocity and spin components at each node. Total speed is at most 35 m/s, individual spin components at most 60 rad/s in magnitude, and maximum early-coordinate residual at most 0.002 m. The inner range is $[1.35, 1.65]^2$. Reported ranges describe the sampled feasible candidates.

Table 6: Ranges of goal-plane crossings compatible with the early positions.

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9122–2.0307	0.3973–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.9075–2.0368	0.3973–0.4043

The largest crossing displacement in the wider scan is 0.1052 m.

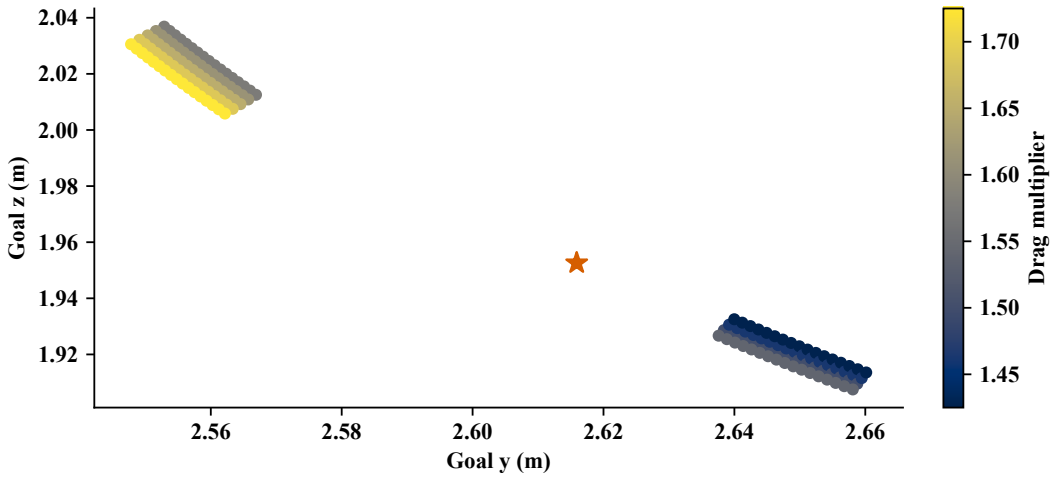


Figure 11: Scanned crossings with early residual at most 2 mm. Colour indicates the drag multiplier.

5.3 Joint perturbations and Monte Carlo simulation

Nine parameters are sampled independently and uniformly over the individual scan ranges, with mass, radius and initial position fixed. Seed 20261108 generates 100000 trajectories; the early positions are allowed to change.

Table 7: Monte Carlo crossing positions and arrival times (100,000 trajectories).

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The crossing mean is $(2.6170, 1.9511)$ m and standard deviations are 0.2455 m and 0.2360 m. Under the ball-centre clearance rule, 95772 trajectories (95.77%) pass fully through the opening and 4228 pass above it.

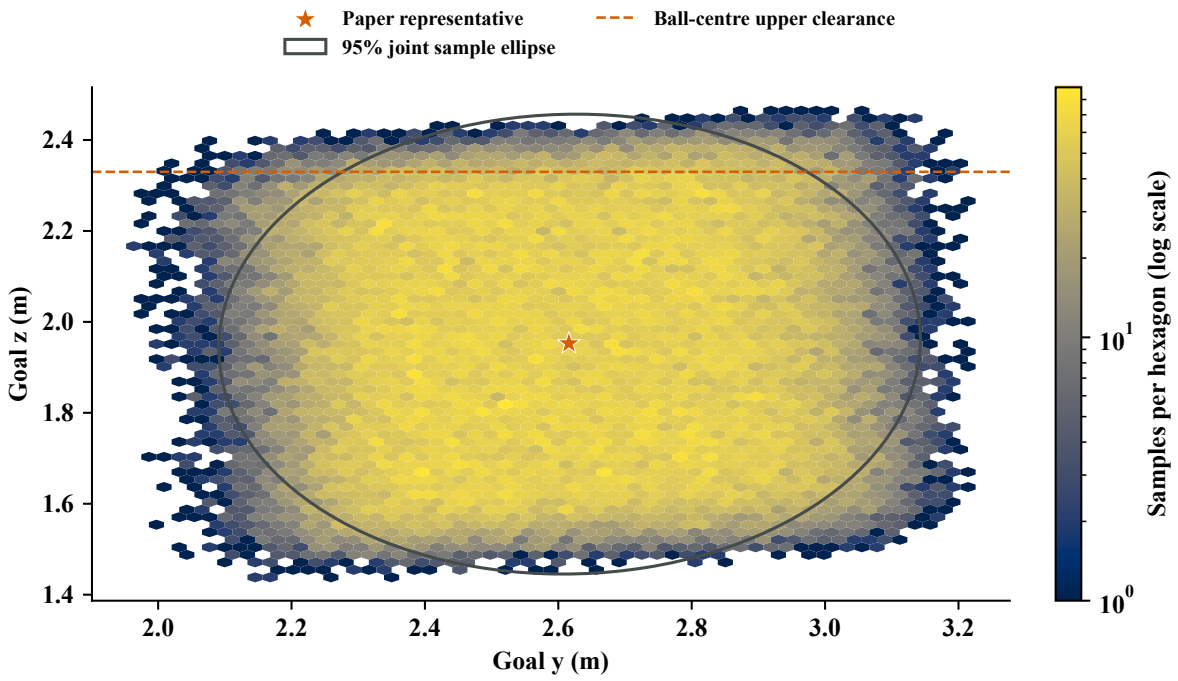


Figure 12: Joint-perturbation crossings; the ellipse covers 95% of simulated samples.

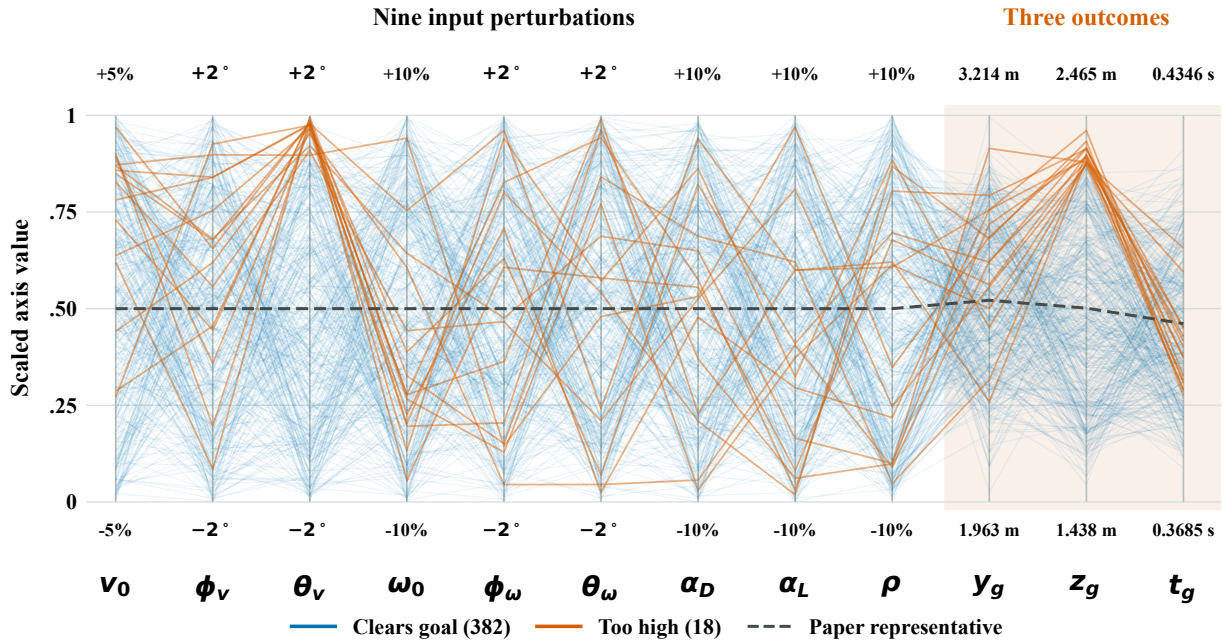


Figure 13: A random display of 400 flights connects the nine inputs to three outputs.

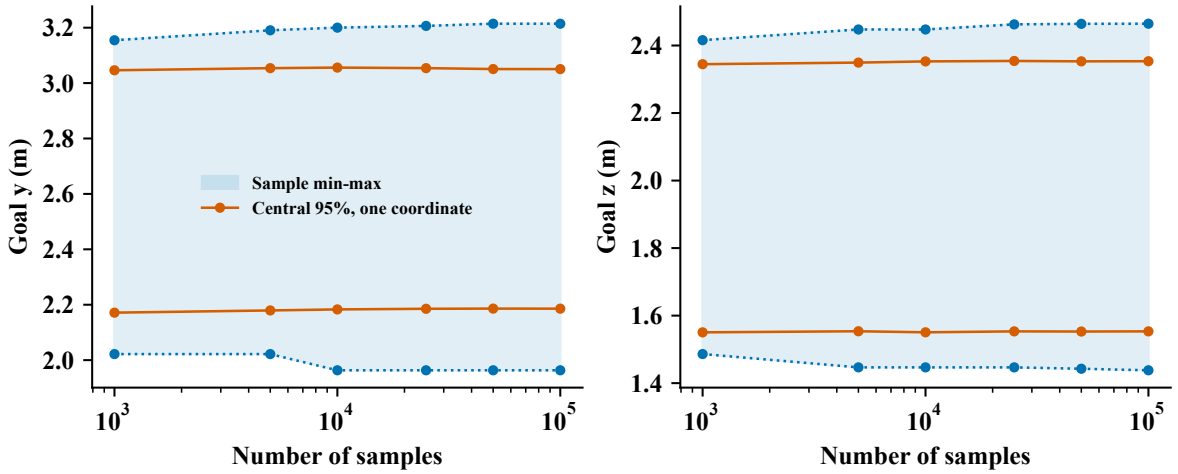


Figure 14: Effect of sample count on the simulated crossing distribution.

Fourth-order Runge–Kutta integration is compared with adaptive integration and a halved step. These ranges and fractions correspond to the stated uniform sampling assumptions.

6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Paudel and Yan [9] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



Figure 15: Illustrative comparison of a paneled football and a smooth sphere, inspired by Paudel and Yan [9]. Their compared bodies had diameter 0.22 m and mass 0.436 kg. The panel geometry shown here is schematic.

Alam et al. [16] measured drag coefficients over a range of Reynolds numbers for six footballs and a smooth sphere. Figure 16 shows the panels and surface textures of these footballs, while Figure 17 compares their drag coefficients. The differences illustrate the influence of ball surface structure on aerodynamic drag.



a) 1966 World Cup replica (18 panels) | Enlarged view of 1966 World Cup replica surface | b) 2002 World Cup replica (32 panels) | Enlarged view of 2002 World Cup Fevernova surface

Fig. 1. Vintage and Adidas Fevernova balls



a) 2006 World Cup replica (14 panels) | Enlarged View of 2006 World Cup Teamgeist II surface | b) 2009 Confederation Cup (14 panels) | Enlarged View of 2009 Confederation Cup Teamgeist III surface

Fig. 2. Adidas Teamgeist 2006 and 2009 balls



a) 2010 World Cup replica (8 panels) | Enlarged view of 2010 World Cup Jabulani surface | b) 2010 Nike T-90 replica (32 panels) | Enlarged view of 2010 Nike T-90 surface

Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

Figure 16: Six footballs and details of their surfaces, reproduced from Figures 1–3 of Alam et al. [16].

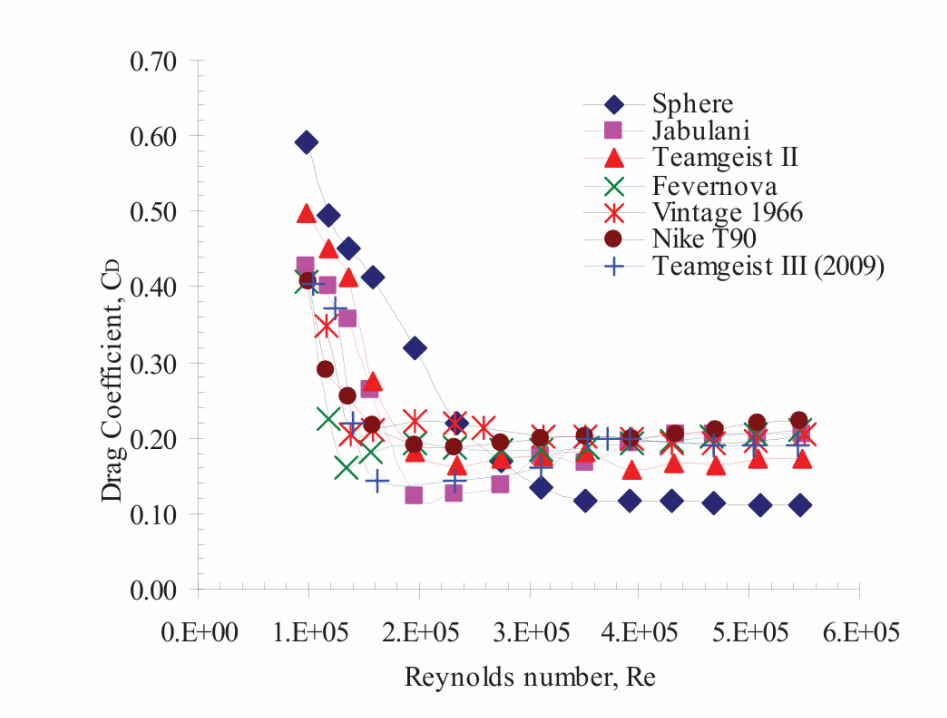


Figure 17: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [16].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Rotation changes Magnus force, so a declining spin can change the continuation of a curved shot. According to Grad et al. [13], Supplementary Section 8.4, and the MuJoCo ellipsoid model [14], the spherical inertial and viscous damping torques are

$$\boldsymbol{Q} = -k_2\omega\boldsymbol{\omega} - k_1\boldsymbol{\omega}, \quad k_2 = \frac{8\pi}{15}\rho C_{\text{angular}} R^5, \quad k_1 = 8\pi\mu R^3. \tag{10}$$

The quadratic term describes high-Reynolds-number angular drag; the linear term is viscous resistance. We use MuJoCo's documented reference value $C_{\text{angular}} = 1.5$ [14]. Defining $q = \rho v^2/2$ and $C_{Q,v} = \|\boldsymbol{Q}\|/(qAR)$ gives

$$C_{Q,v} = \frac{16}{15} C_{\text{angular}} Sp^2 + \frac{32Sp}{Re}, \qquad C_{Q,\omega} = \frac{C_{Q,v}}{Sp^2} = \frac{16}{15} C_{\text{angular}} + \frac{32}{Re\,Sp}. \tag{11}$$

$C_{Q,\omega}$ uses rotating surface speed $R\omega$ in its reference pressure. The factor $1/Sp^2$ converts between the two normalizations. The viscous-to-inertial torque ratio is

$$\frac{Q_{\text{viscous}}}{Q_{\text{inertial}}} = \frac{30}{C_{\text{angular}} Re\, Sp} = \frac{15\nu}{C_{\text{angular}} R^2 \omega}. \tag{12}$$

It is 3.54×10^{-4} at the prescribed curved-shot launch, so the quadratic term dominates. Following the thin-shell inertia used by Maponi and Piombin [8], $I_b = 2mR^2/3$. Adding the high-Reynolds-number torque gives

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = -k_2\omega\boldsymbol{\omega}. \end{cases} \tag{13}$$

The torque opposes spin and removes rotational energy. With this isolated quadratic law, $\omega(t) = \omega_0/[1 + (k_2/I_b)\omega_0 t]$, providing an integration check.

For a prescribed curved shot, with initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with Eq. (13).

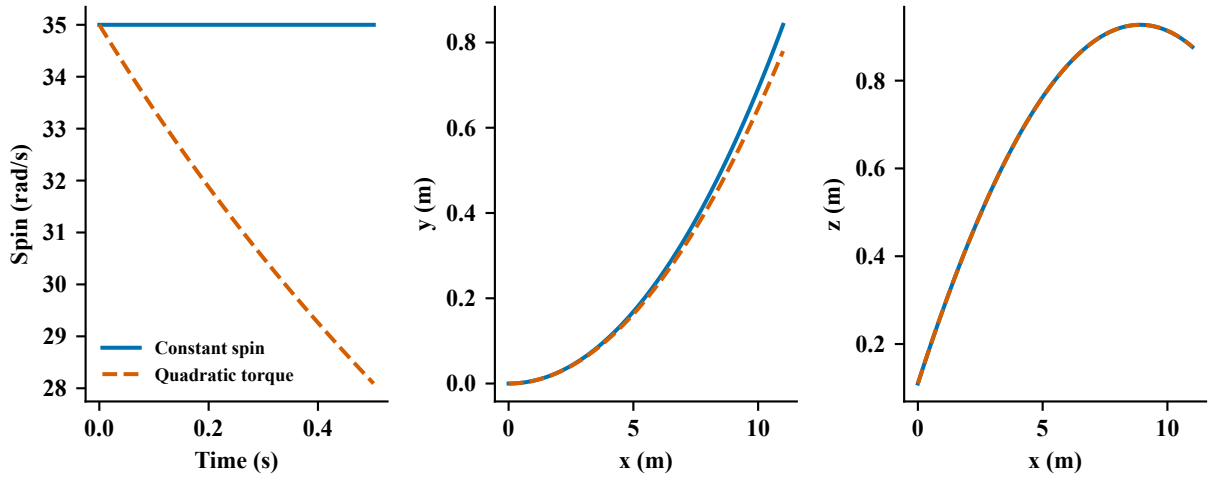


Figure 18: Same-initial-state comparison of constant spin and high-Reynolds-number rotational damping for the prescribed curved shot. The documented angular-drag reference is 1.5, and the ball uses thin-shell inertia.

The damped crossing is $(0.7799, 0.8772)$ m, compared with $(0.8415, 0.8769)$ m at constant spin. The displacement is 0.0616 m (0.560% of 11 m). Spin falls by 19.740%, and crossing time changes by -0.066%. Adding viscous torque shifts the damped crossing by 2.1e-05 m. The short-flight endpoint is less sensitive than spin magnitude to this reference torque.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [8]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{14}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 8 evaluates these rough estimates at the curved-shot launch speed. The listed wind speeds are illustrative conditions for comparison.

Table 8: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The model separates the mechanisms of gravity, drag and Magnus lift. Gravity produces downward curvature, drag reduces speed, and spin redirects the path. We use constant drag and linear-spin lift.

J03WMX and J03WR9 are placement shots and J03WOH is a power shot. Their M4 speeds at the selected start are 29.20, 29.33 and 32.05 m/s, respectively. Drag decreases as the ball slows, while spin direction determines the direction of Magnus force. The four-model comparison shows that fitting the early positions and predicting the goal crossing are distinct tasks: M4 has a 0.113 m crossing error for J03WOH but predicts ground contact before the goal for J03WR9.

Three actual penalties supply distinct early sampling sequences and held-out trajectories. Coordinate conversion places all three in the competition geometry. The 12 inverse outcomes connect the recorded motion to the force assumptions through spatial trajectories and the detailed J03WOH comparison. For the competition-position example in Section 5, a one-degree launch-direction change moves the crossing by the displacement shown in the one-factor table. With the early positions fixed, the explored compatible states shift the crossing by up to 0.1052 m; joint perturbations are quantified by 100,000 Monte Carlo trajectories with fixed ball mass and radius.

8 Model Evaluation

8.1 Physical Interpretation and Reproduction

The four models connect different force assumptions with the recorded flights. Inversion and subsequent prediction use consistent equations. Original source coordinates, selected frames, transformed coordinates, numerical settings and candidate solutions are saved with the results.

8.2 Prediction and Further Development

The real-penalty comparison evaluates how early motion carries into the later path. Bound-active solutions and residuals identify the influence of the chosen coefficient scale and initial-state search. Ball-specific coefficients, direct spin measurements and higher-rate synchronized tracking would improve these connections. Rotational damping and wind can be included when their effect is relevant to the flight considered.

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Reproduction Code

The project contains the original penalty CSV, coordinate conversion, shared inversion, prediction results and figure scripts. The excerpts show the common state equation and force evaluation. Paper and program use the same coefficients and bounds; computation, plotting and bilingual table generation are separate commands.

```
--
    return -factor*w


def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
```

```
--
    return record


def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
```